BANA 6610 - Stats for Business Analytics



As the first step we created our own dataset , thereafter we set the working directory and installed and loaded the required libraries. Next, we read the dataset; the dataset consists of 1030 Observations with 16 Variables.

```
> store<-read.csv("Store.csv")
> store
   Invoice_No      Location       Date Customer.type Gender          Category
1         489 Commerce City 10-03-2023        Member Female     Personal Care
2          82    Louisville 19-01-2023    Non-Member Female           Gadgets
3         359 Commerce City 04-03-2023    Non-Member   Male       Daily Goods
4         515 Commerce City 18-03-2022        Member   Male     Personal Care
5         203 Commerce City 10-07-2022    Non-Member   Male Sports Equipments
6         333    Louisville 22-09-2022    Non-Member   Male           Gadgets
7         597 Commerce City 30-03-2023        Member Female           Gadgets
8         134    Louisville 27-09-2022    Non-Member Female       Daily Goods
9         485 Commerce City 26-05-2022        Member Female     Personal Care
10        725       Greeley 08-08-2022        Member Female         Beverages
11        589       Greeley 27-03-2022        Member Female          Pet Care
12        238       Greeley 06-02-2023        Member   Male           Gadgets
13        528 Commerce City 29-12-2022    Non-Member Female           Gadgets
14        929 Commerce City 25-09-2022    Non-Member   Male         Beverages
15        111 Commerce City 25-11-2022    Non-Member Female     Personal Care
16        535       Greeley 16-03-2023        Member Female Sports Equipments
17        919 Commerce City 13-11-2022        Member Female     Personal Care
18        762 Commerce City 23-01-2023    Non-Member   Male Sports Equipments
19        555 Commerce City 24-02-2023    Non-Member   Male         Beverages
20        556       Greeley 27-08-2022    Non-Member Female       Daily Goods
21        239    Louisville 02-03-2022        Member   Male           Gadgets
22        669       Greeley 06-04-2022    Non-Member   Male     Personal Care
23        261       Greeley 08-07-2022    Non-Member   Male       Daily Goods
24        275 Commerce City 11-11-2022    Non-Member   Male           Gadgets
25        330 Commerce City 02-03-2022        Member   Male Sports Equipments
26        831 Commerce City 16-08-2022        Member Female       Daily Goods
27        580       Greeley 29-08-2022    Non-Member   Male          Pet Care
28        787 Commerce City 01-02-2023    Non-Member Female          Pet Care
29        943       Greeley 14-09-2022    Non-Member Female         Beverages
30        397 Commerce City 19-02-2023    Non-Member   Male     Personal Care
31         88       Greeley 08-05-2022    Non-Member   Male          Pet Care
32          8       Greeley 02-04-2022        Member   Male Sports Equipments
33        345       Greeley 08-05-2022    Non-Member   Male Sports Equipments
```

Next, we found the missing values using is.na and colsums(is.na(store_cleaned_2)). There were no missing values in the dataset.

```
> sum(is.na(store)) # There are no missing values
[1] 0
> colSums(is.na(store))
     Invoice_No        Location            Date   Customer.type          Gender
              0               0               0               0               0
       Category      Price.unit        Quantity         Tax.7.           Total
              0               0               0               0               0
           Time Mode.of.Payment Cost.before.Tax  Gross.Margin..    Gross.Income
              0               0               0               0               0
  Rating.Points
              0
```

We, next went on to clean our dataset, there were no empty rows and columns and named it as store_cleaned_2.

```
> store_1<-clean_names(store)
> store_2<-remove_empty(store_1, which = c("rows","cols"), quiet = FALSE)
No empty rows to remove.
No empty columns to remove.
> store_2
   invoice_no      location       date customer_type gender          category
1         489 Commerce City 10-03-2023        Member Female     Personal Care
2          82    Louisville 19-01-2023    Non-Member Female           Gadgets
3         359 Commerce City 04-03-2023    Non-Member   Male       Daily Goods
4         515 Commerce City 18-03-2022        Member   Male     Personal Care
5         203 Commerce City 10-07-2022    Non-Member   Male Sports Equipments
6         333    Louisville 22-09-2022    Non-Member   Male           Gadgets
7         597 Commerce City 30-03-2023        Member Female           Gadgets
8         134    Louisville 27-09-2022    Non-Member Female       Daily Goods
9         485 Commerce City 26-05-2022        Member Female     Personal Care
10        725       Greeley 08-08-2022        Member Female         Beverages
11        589       Greeley 27-03-2022        Member Female          Pet Care
12        238       Greeley 06-02-2023        Member   Male           Gadgets
13        528 Commerce City 29-12-2022    Non-Member Female           Gadgets
14        929 Commerce City 25-09-2022    Non-Member   Male         Beverages
15        111 Commerce City 25-11-2022    Non-Member Female     Personal Care
16        535       Greeley 16-03-2023        Member Female Sports Equipments
17        919 Commerce City 13-11-2022        Member Female     Personal Care
18        762 Commerce City 23-01-2023    Non-Member   Male Sports Equipments
19        555 Commerce City 24-02-2023    Non-Member   Male         Beverages
20        556       Greeley 27-08-2022    Non-Member Female       Daily Goods
21        239    Louisville 02-03-2022        Member   Male           Gadgets
22        669       Greeley 06-04-2022    Non-Member   Male     Personal Care
23        261       Greeley 08-07-2022    Non-Member   Male       Daily Goods
24        275 Commerce City 11-11-2022    Non-Member   Male           Gadgets
25        330 Commerce City 02-03-2022        Member   Male Sports Equipments
26        831 Commerce City 16-08-2022        Member Female       Daily Goods
27        580       Greeley 29-08-2022    Non-Member   Male          Pet Care
```

```
> store_cleaned_2<-distinct(store_2)
> store_cleaned_2
   invoice_no      location       date customer_type gender          category
1         489 Commerce City 10-03-2023        Member Female     Personal Care
2          82    Louisville 19-01-2023    Non-Member Female           Gadgets
3         359 Commerce City 04-03-2023    Non-Member   Male       Daily Goods
4         515 Commerce City 18-03-2022        Member   Male     Personal Care
5         203 Commerce City 10-07-2022    Non-Member   Male Sports Equipments
6         333    Louisville 22-09-2022    Non-Member   Male           Gadgets
7         597 Commerce City 30-03-2023        Member Female           Gadgets
8         134    Louisville 27-09-2022    Non-Member Female       Daily Goods
9         485 Commerce City 26-05-2022        Member Female     Personal Care
10        725       Greeley 08-08-2022        Member Female         Beverages
11        589       Greeley 27-03-2022        Member Female          Pet Care
12        238       Greeley 06-02-2023        Member   Male           Gadgets
13        528 Commerce City 29-12-2022    Non-Member Female           Gadgets
14        929 Commerce City 25-09-2022    Non-Member   Male         Beverages
15        111 Commerce City 25-11-2022    Non-Member Female     Personal Care
16        535       Greeley 16-03-2023        Member Female Sports Equipments
17        919 Commerce City 13-11-2022        Member Female     Personal Care
18        762 Commerce City 23-01-2023    Non-Member   Male Sports Equipments
19        555 Commerce City 24-02-2023    Non-Member   Male         Beverages
20        556       Greeley 27-08-2022    Non-Member Female       Daily Goods
21        239    Louisville 02-03-2022        Member   Male           Gadgets
22        669       Greeley 06-04-2022    Non-Member   Male     Personal Care
23        261       Greeley 08-07-2022    Non-Member   Male       Daily Goods
24        275 Commerce City 11-11-2022    Non-Member   Male           Gadgets
25        330 Commerce City 02-03-2022        Member   Male Sports Equipments
26        831 Commerce City 16-08-2022        Member Female       Daily Goods
27        580       Greeley 29-08-2022    Non-Member   Male          Pet Care
28        787 Commerce City 01-02-2023    Non-Member Female          Pet Care
29        943       Greeley 14-09-2022    Non-Member Female         Beverages
30        397 Commerce City 19-02-2023    Non-Member   Male     Personal Care
31         88       Greeley 08-05-2022    Non-Member   Male          Pet Care
```

Further, summary, str, class, head and tail command on the cleaned dataset (store_cleaned_2); summary command is used to describe the 1st 2nd 3rd quartile along with upper and lower whiskers, str command is used to check what are the datatypes of the variables, class command tells us the datatype (dataframe), head command is used to get the glimpse of first 5 rows with all columns and tail command is used to get the glimpse of last 5 rows with all columns.

```
> summary(store_cleaned_2)
   invoice_no       location            date          customer_type
 Min.   :   2.0   Length:1030        Length:1030        Length:1030
 1st Qu.: 406.8   Class :character   Class :character   Class :character
 Median : 811.5   Mode  :character   Mode  :character   Mode  :character
 Mean   :1229.8
 3rd Qu.:1767.8
 Max.   :9643.0
    gender            category          price_unit       quantity
 Length:1030        Length:1030        Min.   : 11.23   Min.   : 3.000
 Class :character   Class :character   1st Qu.: 34.46   1st Qu.: 5.000
 Mode  :character   Mode  :character   Median : 56.38   Median : 7.000
                                       Mean   : 56.95   Mean   : 7.485
                                       3rd Qu.: 79.09   3rd Qu.:10.000
                                       Max.   :101.11   Max.   :12.000
     tax_7             total             time           mode_of_payment
 Min.   : 2.377   Min.   :  36.34   Length:1030        Length:1030
 1st Qu.:15.358   1st Qu.: 234.76   Class :character   Class :character
 Median :25.682   Median : 392.57   Mode  :character   Mode  :character
 Mean   :29.896   Mean   : 456.98
 3rd Qu.:41.773   3rd Qu.: 638.53
 Max.   :84.378   Max.   :1289.78
 cost_before_tax    gross_margin  gross_income    rating_points
 Min.   :  33.96   Min.   :5.7   Min.   : 2.071   Min.   :1.100
 1st Qu.: 219.41   1st Qu.:5.7   1st Qu.:13.382   1st Qu.:3.400
 Median : 366.88   Median :5.7   Median :22.376   Median :5.600
 Mean   : 427.08   Mean   :5.7   Mean   :26.048   Mean   :5.597
 3rd Qu.: 596.76   3rd Qu.:5.7   3rd Qu.:36.396   3rd Qu.:7.900
 Max.   :1205.40   Max.   :5.7   Max.   :73.517   Max.   :9.900
> str(store_cleaned_2)
'data.frame':   1030 obs. of  16 variables:
 $ invoice_no     : int  489 82 359 515 203 333 597 134 485 725 ...
 $ location       : chr  "Commerce City" "Louisville" "Commerce City" "Commerce City" ...
 $ date           : chr  "10-03-2023" "19-01-2023" "04-03-2023" "18-03-2022" ...
 $ customer_type  : chr  "Member" "Non-Member" "Non-Member" "Member" ...
 $ gender         : chr  "Female" "Female" "Male" "Male" ...
 $ category       : chr  "Personal Care" "Gadgets" "Daily Goods" "Personal Care" ...
 $ price_unit     : num  75.8 16.4 47.5 59.4 87.5 ...
 $ quantity       : int  9 7 9 10 9 9 8 12 4 5 ...
 $ tax_7          : num  47.78 8.05 29.91 41.56 55.1 ...
 $ total          : num  730 123 457 635 842 ...
 $ time           : chr  "13:19" "23:48" "05:06" "21:15" ...
 $ mode_of_payment: chr  "Ewallet" "Cash" "Card(Credit/Debit)" "Ewallet" ...
 $ cost_before_tax: num  683 115 427 594 787 ...
 $ gross_margin   : num  5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 ...
 $ gross_income   : num  41.63 7.01 26.06 36.21 48.01 ...
 $ rating_points  : num  5.2 4.9 3 9.5 7.6 2.7 9.7 2.8 2 8.6 ...
```

```
> head(store_cleaned_2)
  invoice_no      location       date customer_type gender          category
1        489 Commerce City 10-03-2023        Member Female     Personal Care
2         82    Louisville 19-01-2023    Non-Member Female           Gadgets
3        359 Commerce City 04-03-2023    Non-Member   Male       Daily Goods
4        515 Commerce City 18-03-2022        Member   Male     Personal Care
5        203 Commerce City 10-07-2022    Non-Member   Male Sports Equipments
6        333    Louisville 22-09-2022    Non-Member   Male           Gadgets
  price_unit quantity   tax_7    total  time    mode_of_payment cost_before_tax
1      75.84        9 47.7792 730.3392 13:19            Ewallet          682.56
2      16.43        7  8.0507 123.0607 23:48               Cash          115.01
3      47.48        9 29.9124 457.2324 05:06 Card(Credit/Debit)          427.32
4      59.37       10 41.5590 635.2590 21:15            Ewallet          593.70
5      87.46        9 55.0998 842.2398 01:42            Ewallet          787.14
6      86.54        9 54.5202 833.3802 09:04            Ewallet          778.86
  gross_margin gross_income rating_points
1          5.7     41.62933           5.2
2          5.7      7.01446           4.9
3          5.7     26.06225           3.0
4          5.7     36.20976           9.5
5          5.7     48.00767           7.6
6          5.7     47.50267           2.7
> tail(store_cleaned_2)
     invoice_no      location       date customer_type gender          category
1025       5296    Louisville 07-11-2022    Non-Member   Male     Personal Care
1026       5146       Greeley 16-04-2022    Non-Member   Male          Pet Care
1027       5211       Greeley 29-10-2022    Non-Member   Male Sports Equipments
1028       5879 Commerce City 02-10-2022        Member   Male     Personal Care
1029       9643 Commerce City 04-10-2022    Non-Member Female         Beverages
1030       7654 Commerce City 20-11-2022    Non-Member Female          Pet Care
     price_unit quantity   tax_7     total  time    mode_of_payment cost_before_tax
1025      82.86        8 46.4016  709.2816 20:41 Card(Credit/Debit)          662.88
1026      92.56        7 45.3544  693.2744 01:42            Ewallet          647.92
1027      40.36        6 16.9512  259.1112 14:10               Cash          242.16
1028      61.01        4 17.0828  261.1228 13:53               Cash          244.04
1029      55.51       12 46.6284  712.7484 15:52            Ewallet          666.12
1030      99.24       11 76.4148 1168.0548 22:23 Card(Credit/Debit)         1091.64
     gross_margin gross_income rating_points
1025          5.7     40.42905           4.5
1026          5.7     39.51664           8.6
1027          5.7     14.76934           7.5
1028          5.7     14.88400           2.7
1029          5.7     40.62666           5.6
1030          5.7     66.57912           4.5
> class(store_cleaned_2)
[1] "data.frame"
```

Factorization for the character variables of the dataset, factorization is generally done to implement the character or categorical variables rightly for model building.

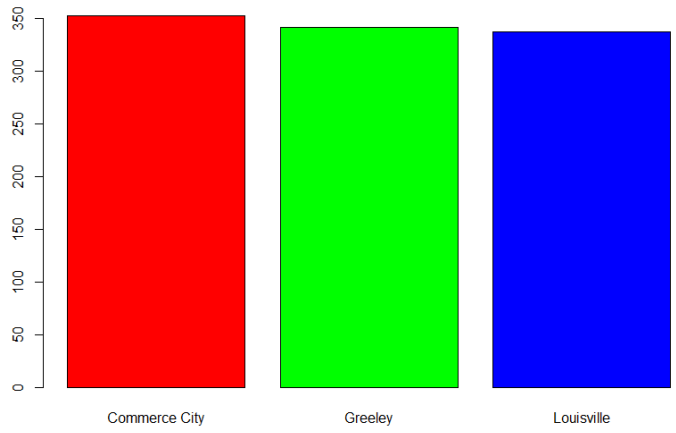
```
> store_cleaned_2$location<-as.factor(store_cleaned_2$location)
> store_cleaned_2$customer_type<-as.factor(store_cleaned_2$customer_type)
> store_cleaned_2$gender<-as.factor(store_cleaned_2$gender)
> store_cleaned_2$category<-as.factor(store_cleaned_2$category)
> store_cleaned_2$mode_of_payment<-as.factor(store_cleaned_2$mode_of_payment)
```

```
> str(store_cleaned_2)
'data.frame':   1030 obs. of  16 variables:
 $ invoice_no     : int  489 82 359 515 203 333 597 134 485 725 ...
 $ location       : Factor w/ 3 levels "Commerce City",..: 1 3 1 1 1 3 1 3 1 2 ...
 $ date           : chr  "10-03-2023" "19-01-2023" "04-03-2023" "18-03-2022" ...
 $ customer_type  : Factor w/ 2 levels "Member","Non-Member": 1 2 2 1 2 2 1 2 1 1 ...
 $ gender         : Factor w/ 2 levels "Female","Male": 1 1 2 2 2 2 1 1 1 1 ...
 $ category       : Factor w/ 6 levels "Beverages","Daily Goods",..: 4 3 2 4 6 3 3 2 4 1
 ...
 $ price_unit     : num  75.8 16.4 47.5 59.4 87.5 ...
 $ quantity       : int  9 7 9 10 9 9 8 12 4 5 ...
 $ tax_7          : num  47.78 8.05 29.91 41.56 55.1 ...
 $ total          : num  730 123 457 635 842 ...
 $ time           : chr  "13:19" "23:48" "05:06" "21:15" ...
 $ mode_of_payment: Factor w/ 3 levels "Card(Credit/Debit)",..: 3 2 1 3 3 3 3 3 1 1 ...
 $ cost_before_tax: num  683 115 427 594 787 ...
 $ gross_margin   : num  5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 5.7 ...
 $ gross_income   : num  41.63 7.01 26.06 36.21 48.01 ...
 $ rating_points  : num  5.2 4.9 3 9.5 7.6 2.7 9.7 2.8 2 8.6 ...
```

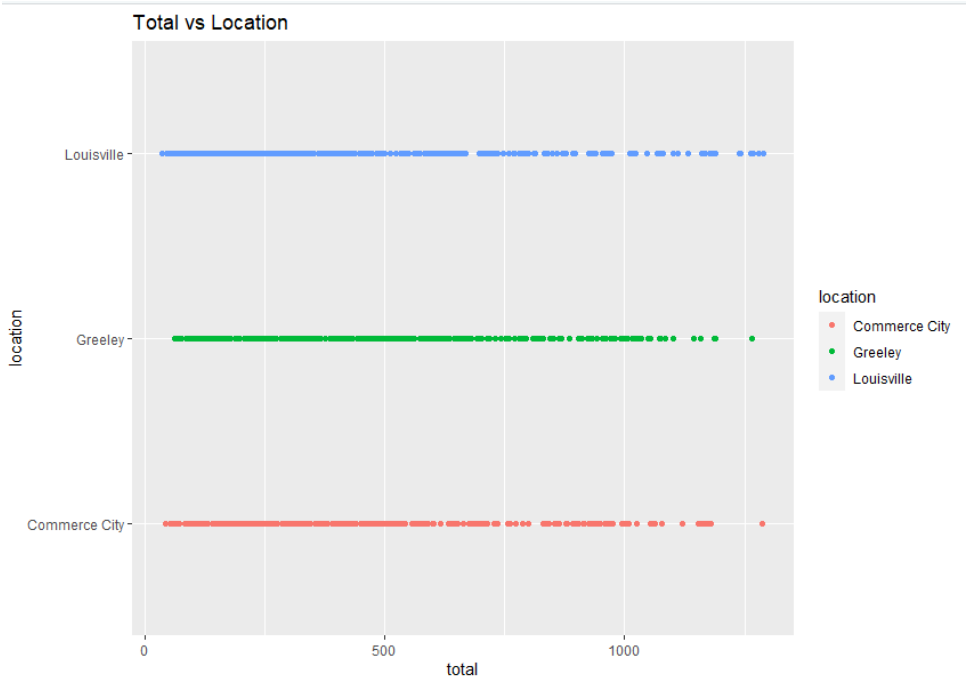
There are a total of 3 Locations in the dataset based on which all the sales are determined; Commerce City, Greeley and Louisville are the three locations.

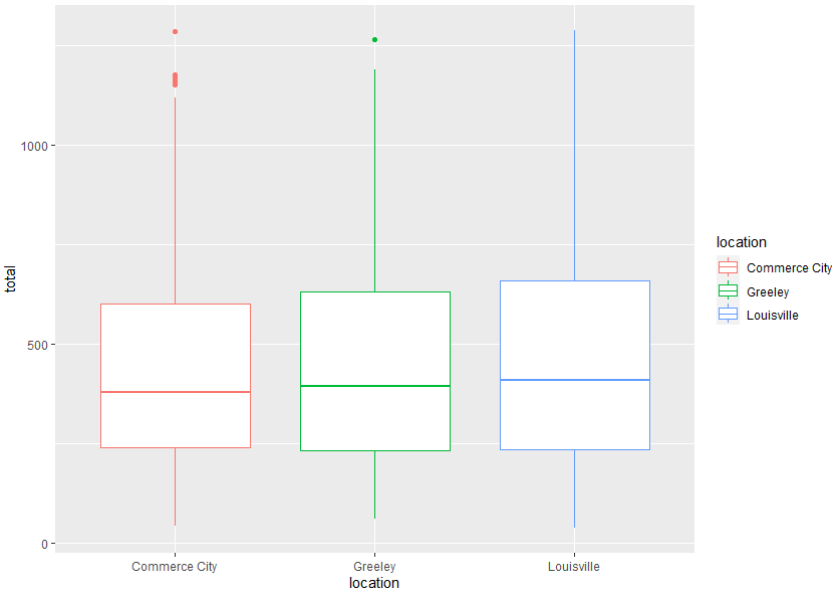
```
> location_counts<- table(store_cleaned_2$location)
>
> print(location_counts)

Commerce City       Greeley     Louisville
          352           341            337
>
> colours<-c("red","green","blue")
>
> barplot(location_counts, col = c("red","green","blue"))
```



Below is the First and Second data visualization we created which is a scatter plot and boxplots for Total Sales vs Location. As we observe Commerce City has the highest sales with 352 followed by Greeley with 341 next comes Louisville with 337. All the boxplots for the locations are almost normally distributed and Commerce City and Greeley have a few outliers.

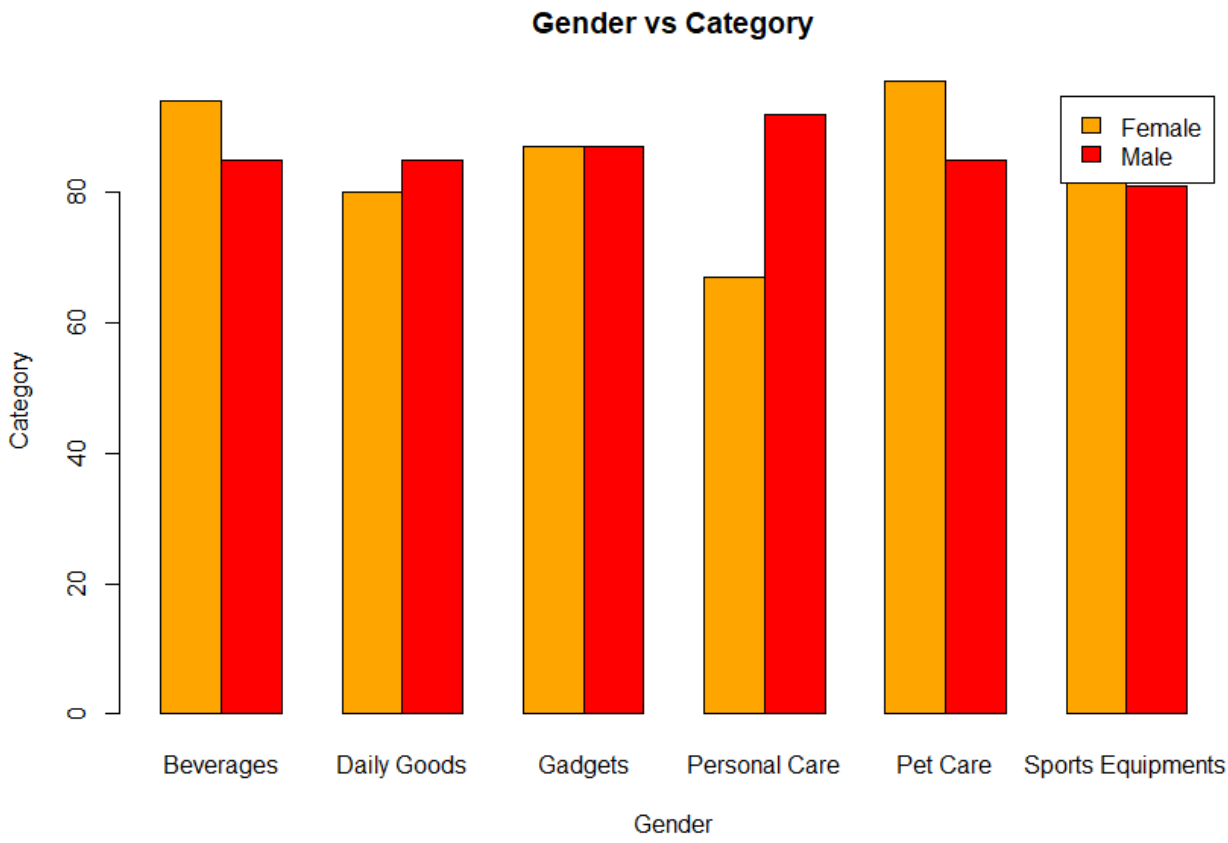




The Third data visualization we created is a barplot for gender and category, as we observe there are 6 categories of products; Beverages, Daily Goods, Gadgets, Personal Care, Pet Care and Sports Equipment and two genders. Further, Females are more inclined in buying the Beverages, Pet Care and Sports Equipment, whereas Males often tend to buy Personal Care products and Daily Goods. Gadgets is only type of product where both Males and Females are equal.

```
> # Barplot for gender and category
> counts<-table(store_cleaned_2$gender,store_cleaned_2$category);counts

        Beverages Daily Goods Gadgets Personal Care Pet Care Sports Equipments
  Female        94          80      87            67       97                90
  Male          85          85      87            92       85                81
> rownames(counts)
[1] "Female" "Male"
> barplot(counts,main = "Gender vs Category",xlab = 'Gender',ylab = 'Category',
+         col = c("Orange","Red"), legend=rownames(counts),beside = T)
```

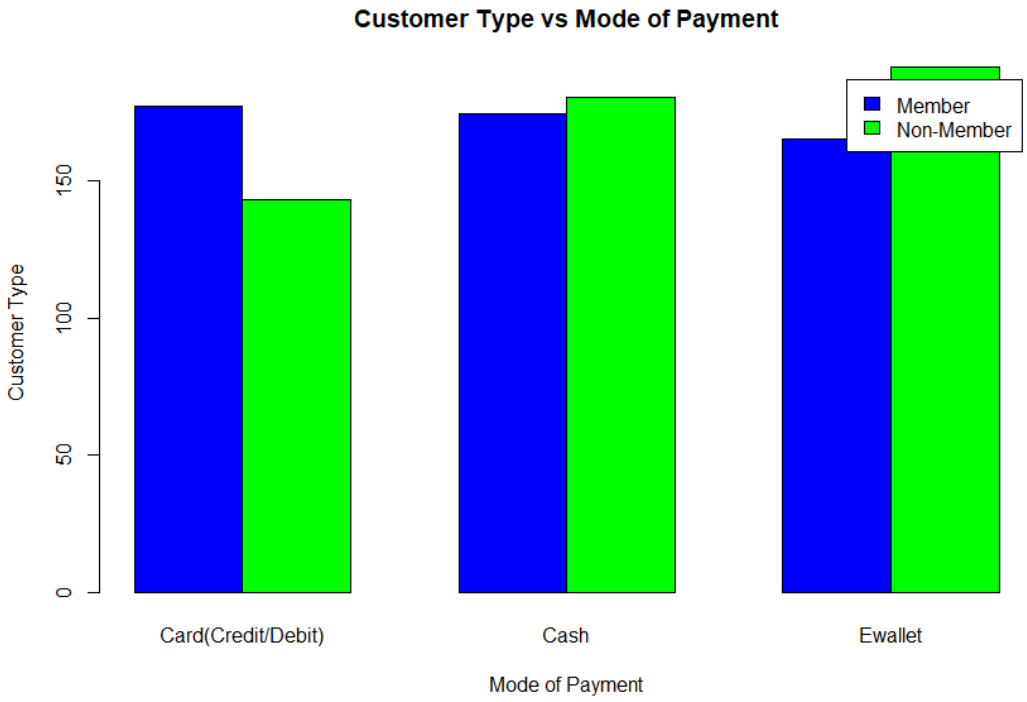


The Fourth data visualization we created is a barplot for customer type and mode of payment, as we observe there are 3 types of payment methods; Card (Credit/Debit), Cash and Ewallet (New

Trend). Further, there are two types of customers; Member and Non-member. Irrespective of the customer type most customers prefer to buy their goods via E-wallet, followed by Cash and Card.

```
> counts_1<-table(store_cleaned_2$customer_type,store_cleaned_2$mode_of_payment);counts_1

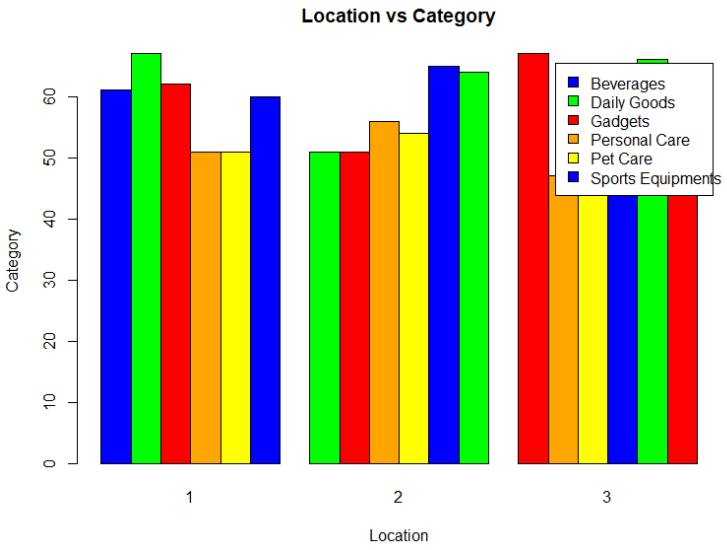
             Card(Credit/Debit) Cash Ewallet
  Member                    177  174     165
  Non-Member                143  180     191
>
> rownames(counts_1)
[1] "Member"     "Non-Member"
>
> barplot(counts_1,main = "Customer Type vs Mode of Payment",xlab = 'Mode of Payment',ylab
 = 'Customer Type',
+         col = c("Blue","Green"), legend=rownames(counts_1),beside = T)
```



The Fifth data visualization we created is a barplot for category and location, 6 categories of products; Beverages, Daily Goods, Gadgets, Personal Care, Pet Care and Sports Equipment. Further, there are 3 locations; Commerce City (1), Greeley (2) and Louisville (3). As, we observe Daily Goods in location 1 has the highest sales and least in location 2, Beverages in location 3 has the highest sales and least in location 2 and Sport Equipment has the highest sales in location 2 and least in location 3.

```
> counts_2<-table(store_cleaned_2$category,store_cleaned_2$location);counts_2

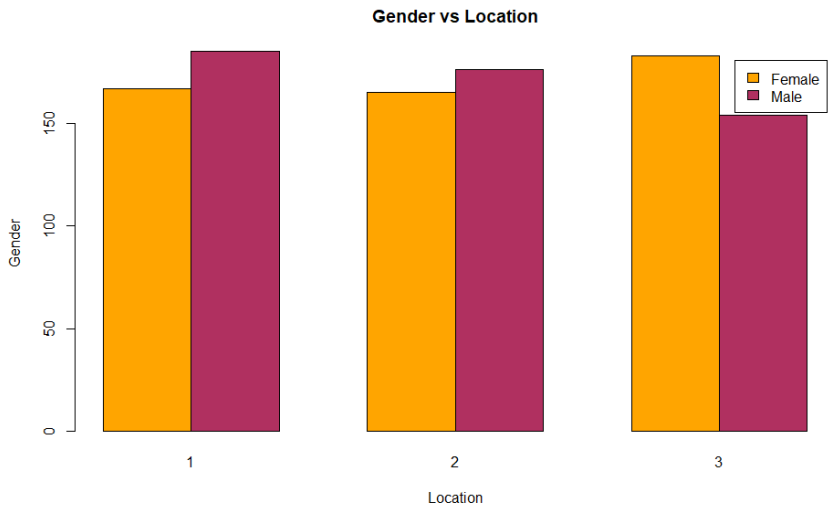
                    1  2  3
  Beverages        61 51 67
  Daily Goods      67 51 47
  Gadgets          62 56 56
  Personal Care    51 54 54
  Pet Care         51 65 66
  Sports Equipments 60 64 47
>
> rownames(counts_2)
[1] "Beverages"         "Daily Goods"       "Gadgets"           "Personal Care"
[5] "Pet Care"          "Sports Equipments"
>
> barplot(counts_2,main = "Location vs Category",xlab = 'Location',ylab = 'Category',
+         col = c("Blue","Green","Red","Orange","Yellow"), legend=rownames(counts_2),beside = T)
```



The Sixth data visualization we created is a barplot for gender and location. Gender; Male and Female. Further, there are 3 locations; Commerce City (1), Greeley (2) and Louisville (3). We can conclude a few things that most of males visit the location 1 and least at location 2 and most of the females visit the location 3 and least at location 3.

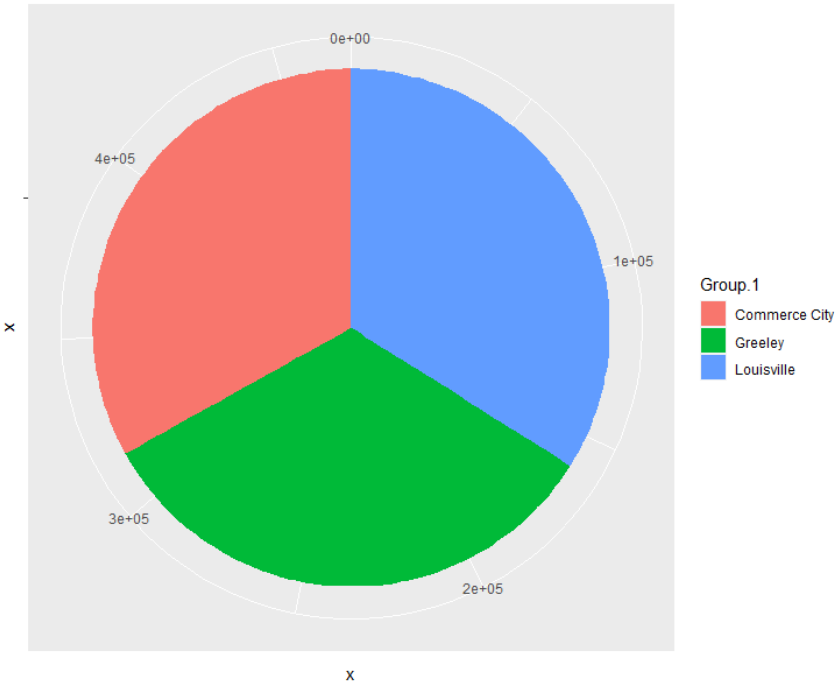
```
> # Barplot for gender and location
> counts_3<-table(store_cleaned_2$gender,store_cleaned_2$location);counts_3

           1   2   3
  Female 167 165 183
  Male   185 176 154
> rownames(counts_3)
[1] "Female" "Male"
> barplot(counts_3,main = "Gender vs Location",xlab = 'Gender',ylab = 'Location',
+         col = c("Orange","Maroon"), legend=rownames(counts_3),beside = T)
> barplot(counts_3,main = "Gender vs Location",xlab = 'Location',ylab = 'Gender',
+         col = c("Orange","Maroon"), legend=rownames(counts_3),beside = T)
```



The Seventh data visualization is a pie chart which depicts the total percentage of total sales for each Location, as we observe Louisville take up the maximum percentage with 33.94% followed by Commerce City with 33.07% then comes Greeley with 32.99%. A pie chart represents the data in the form of a “pie”; a circular graph. Each pie chart depicts different levels or portions of the data. The whole pie chart is 100%.

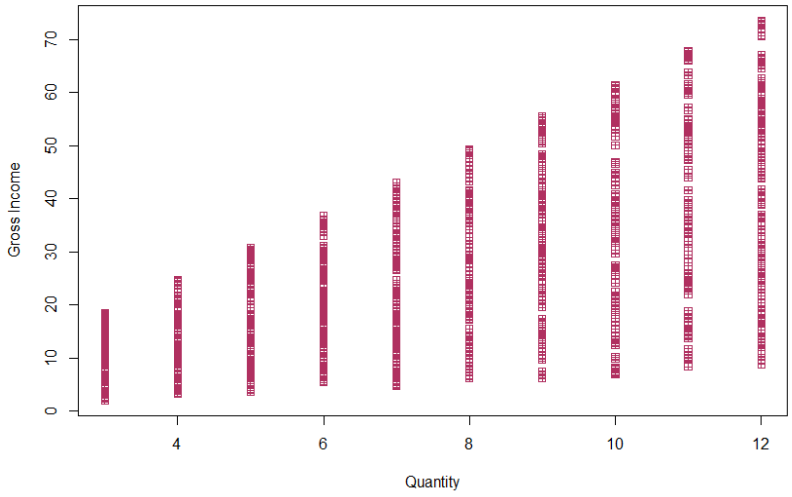
The pie chart is followed by a table which displays the percentage of total sales.

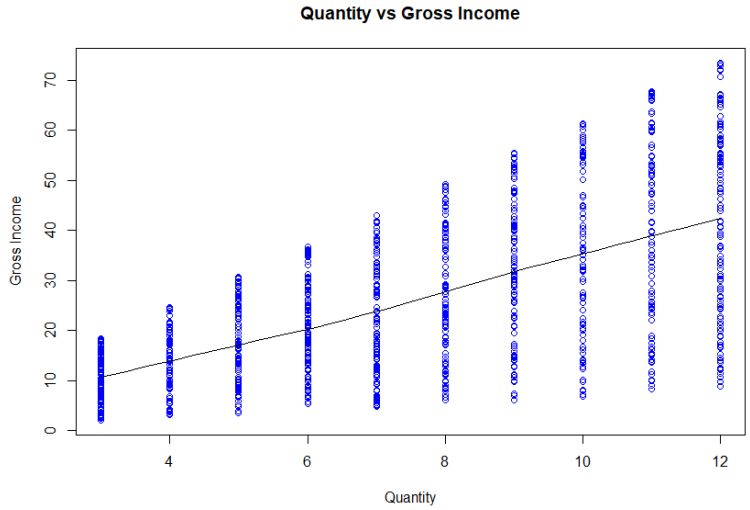


Row Labels	Sum of Total	Sum of Total2
Commerce City	155658.9348	33.07%
Greeley	155282.8298	32.99%
Louisville	159744.1841	33.94%
(blank)		0.00%
Grand Total	**470685.9487**	**100.00%**

The Eight and Last data visualization are two scatter plots; one with normal scatter plot and the other with scatter plot with the trend line. The scatter plot is plotted for Quantity vs Gross Income, there exists a positive relationship between the two, which further means that the gross income increases when the quantity is high.

```
> cor(store_cleaned_2$quantity,store_cleaned_2$gross_income)
[1] 0.6210297
```





Multiple Linear Model

A multiple linear model is a statistical model used to find and analyse the linear relationship between one dependent variables and multiple independent variables. In simple words it's a regression model which includes two or more predictors.

We first created a SISO Model (Single Input with Single Output) and we got R Square value of 100% and our Adjusted R Square value of same 100%.

```
> siso<-lm(total~gross_income,data = store_cleaned_2);summary(siso)

Call:
lm(formula = total ~ gross_income, data = store_cleaned_2)

Residuals:
       Min         1Q     Median         3Q        Max
-2.061e-13 -4.800e-14 -2.610e-14  2.300e-15  1.261e-11

Coefficients:
              Estimate Std. Error   t value Pr(>|t|)
(Intercept)  1.020e-12  3.120e-14 3.270e+01   <2e-16 ***
gross_income 1.754e+01  1.010e-15 1.736e+16   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.376e-13 on 1028 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 3.015e+32 on 1 and 1028 DF,  p-value: < 2.2e-16
```

Next, we went on creating a series of MISO Models (Multiple Inputs with a Single Output), we finally selected two MISO models one with the Total (Dependent Variable) and two Independent Variables (Price unit and Quantity). Our next MISO model has the same Dependent Variable with two different Independent Variables (Price unit and Gross Income)

```
> miso_main<-lm(total~price_unit+quantity,data=store_cleaned_2);summary(miso_main)

Call:
lm(formula = total ~ price_unit + quantity, data = store_cleaned_2)

Residuals:
     Min       1Q   Median       3Q      Max
-216.762  -47.602    0.573   46.481  221.851

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -460.52100    8.97227  -51.33   <2e-16 ***
price_unit     8.05712    0.09758   82.57   <2e-16 ***
quantity      61.26698    0.88323   69.37   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 82.61 on 1027 degrees of freedom
Multiple R-squared:  0.9196,    Adjusted R-squared:  0.9194
F-statistic:  5872 on 2 and 1027 DF,  p-value: < 2.2e-16
```

```
> miso_2<-lm(total~price_unit+quantity+location,data=store_cleaned_2);summary(miso_2)

Call:
lm(formula = total ~ price_unit + quantity + location, data = store_cleaned_2)

Residuals:
    Min      1Q  Median      3Q     Max
-216.12  -47.22    1.92   46.65  217.47

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)        -463.52787    9.57646 -48.403   <2e-16 ***
price_unit            8.05354    0.09765  82.472   <2e-16 ***
quantity             61.24708    0.88363  69.313   <2e-16 ***
locationGreeley       2.75190    6.27970   0.438    0.661
locationLouisville    7.48474    6.30182   1.188    0.235
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 82.63 on 1025 degrees of freedom
Multiple R-squared:  0.9197,    Adjusted R-squared:  0.9194
F-statistic:  2935 on 4 and 1025 DF,  p-value: < 2.2e-16



> miso_3<-lm(total~price_unit+quantity+time,data=store_cleaned_2);summary(miso_3)

Call:
lm(formula = total ~ price_unit + quantity + time, data = store_cleaned_2)

Residuals:
     Min       1Q   Median       3Q      Max
-216.029  -44.238   -1.255   46.227  227.443

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -438.1882    21.6089 -20.278  < 2e-16 ***
price_unit     7.9888     0.1018  78.494  < 2e-16 ***
quantity      61.1248     0.9264  65.982  < 2e-16 ***
time00:17    -46.0708    33.0001  -1.396  0.16302
time00:34    -30.0028    28.3181  -1.059  0.28965
time00:51    -14.5481    30.4475  -0.478  0.63290
time01:08     -5.4885    30.4403  -0.180  0.85695
time01:25    -15.2424    37.0850  -0.411  0.68116
time01:42    -15.1688    30.4945  -0.497  0.61900
time01:59    -17.7556    31.9584  -0.556  0.57863
time02:16    -22.9669    31.9744  -0.718  0.47276
time02:33     -9.0567    29.2417  -0.310  0.75684
time02:50    -29.4707    39.2336  -0.751  0.45275
time03:07     -5.7984    28.8847  -0.201  0.84094
time03:24      0.2343    35.4120   0.007  0.99472
time03:41      5.4064    32.9213   0.164  0.86959
time03:58     18.5984    28.8876   0.644  0.51985
time04:15    -30.1985    32.0240  -0.943  0.34592
time04:32    -50.5214    37.0677  -1.363  0.17322
```

```
time15:35    -19.7763    32.9917  -0.599  0.54903
time15:52    -10.0719    31.9737  -0.315  0.75283
time16:09    -35.9012    29.8636  -1.202  0.22960
time16:26    -38.7616    30.4366  -1.274  0.20315
time16:43    -45.6664    34.0477  -1.341  0.18016
time17:00     -3.0038    34.1429  -0.088  0.92991
time17:17    -12.4631    34.1056  -0.365  0.71487
time17:34    -68.3164    31.9603  -2.138  0.03281 *
time17:51    -38.5916    29.8491  -1.293  0.19637
time18:08      1.1460    33.0082   0.035  0.97231
time18:25    -36.0856    31.1247  -1.159  0.24659
time18:42    -11.8931    28.7730  -0.413  0.67945
time18:59     13.2808    32.9272   0.403  0.68679
time19:16    -30.8439    31.1361  -0.991  0.32213
time19:33    -10.9479    28.7748  -0.380  0.70368
time19:50    -36.4973    34.0728  -1.071  0.28437
time20:07    -14.6161    29.7953  -0.491  0.62386
time20:24     37.7206    30.4183   1.240  0.21526
time20:41    -51.2699    29.2509  -1.753  0.07997 .
time20:58    -34.7362    31.9387  -1.088  0.27705
time21:15    -22.6481    31.1827  -0.726  0.46783
time21:32    -19.5151    27.5941  -0.707  0.47961
time21:49     11.6197    31.1250   0.373  0.70899
time22:06    -49.0134    35.4680  -1.382  0.16733
time22:23     41.3934    30.4232   1.361  0.17397
time22:40     -0.5761    29.8302  -0.019  0.98460
time22:57     53.1556    37.1318   1.432  0.15261
time23:14     -7.7285    27.9339  -0.277  0.78209
time23:31    -40.6350    30.4142  -1.336  0.18185
time23:48      3.7685    39.2146   0.096  0.92346
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 82.54 on 943 degrees of freedom
Multiple R-squared:  0.9263,    Adjusted R-squared:  0.9196
F-statistic: 137.8 on 86 and 943 DF,  p-value: < 2.2e-16
```

```
> miso_5<-lm(total~price_unit+gross_income+rating_points,data=store_cleaned_2);summary(mis
o_5)

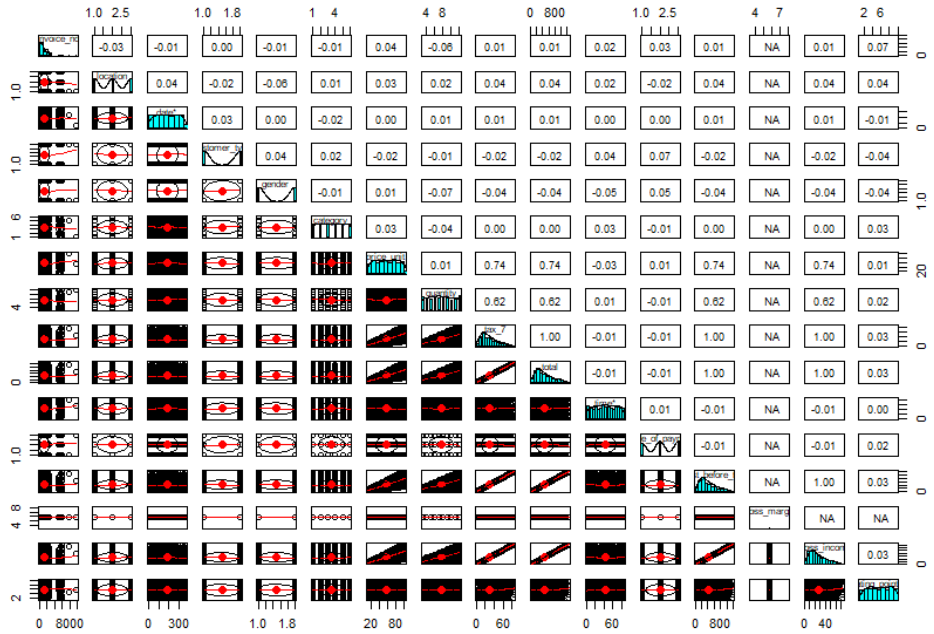
Call:
lm(formula = total ~ price_unit + gross_income + rating_points,
    data = store_cleaned_2)

Residuals:
       Min         1Q     Median         3Q        Max
-2.379e-13 -3.770e-14 -1.320e-14  1.300e-14  1.196e-11

Coefficients:
               Estimate Std. Error    t value Pr(>|t|)
(Intercept)    5.326e-13  3.841e-14  1.387e+01  < 2e-16 ***
price_unit    -5.371e-15  6.724e-16 -7.988e+00 3.65e-15 ***
gross_income   1.754e+01  1.070e-15  1.639e+16  < 2e-16 ***
rating_points -8.468e-16  4.648e-15 -1.820e-01    0.855
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.849e-13 on 1026 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 1.961e+32 on 3 and 1026 DF,  p-value: < 2.2e-16
```

We, then plotted a pairs plot for the cleaned dataset (store_cleaned_2), we use the pairs plot to check relationship of each Independent Variable for the Dependent Variable and to understand the clusters. As we observe, most of the plots are right skewed and a few with a different one.



Advanced Models

First, we created a Linear Regression Model for the main miso model (miso_main).

When looking at the output of a statistical analysis we used a Gaussian (normal) distribution to model the response variable. The dispersion parameter for the Gaussian family is typically the variance of the response variable. In this case, the dispersion parameter is given as 6824.636.

The null deviance and residual deviance are measures of the goodness of fit of the model. The null deviance represents the total variability in the response variable when no predictors are included in the model. The residual deviance represents the unexplained variability in the response variable after accounting for the predictors in the model. In this case, the null deviance is 87151479 and the residual deviance is 7008901. The difference between the two is used to check the accuracy.

The AIC (Akaike Information Criterion) is a measure of the relative quality of a statistical model for a given set of data. It takes into account both the goodness of fit of the model and the complexity of the model (i.e., the number of parameters). A lower AIC value indicates a better model fit. In this case, the AIC value is 12021, which suggests a relatively good fit of the model to the data.

```
> linear_regression<-glm(total~price_unit+quantity,data=store_cleaned_2);summary(linear_regression)

Call:
glm(formula = total ~ price_unit + quantity, data = store_cleaned_2)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-216.762   -47.602     0.573    46.481   221.851

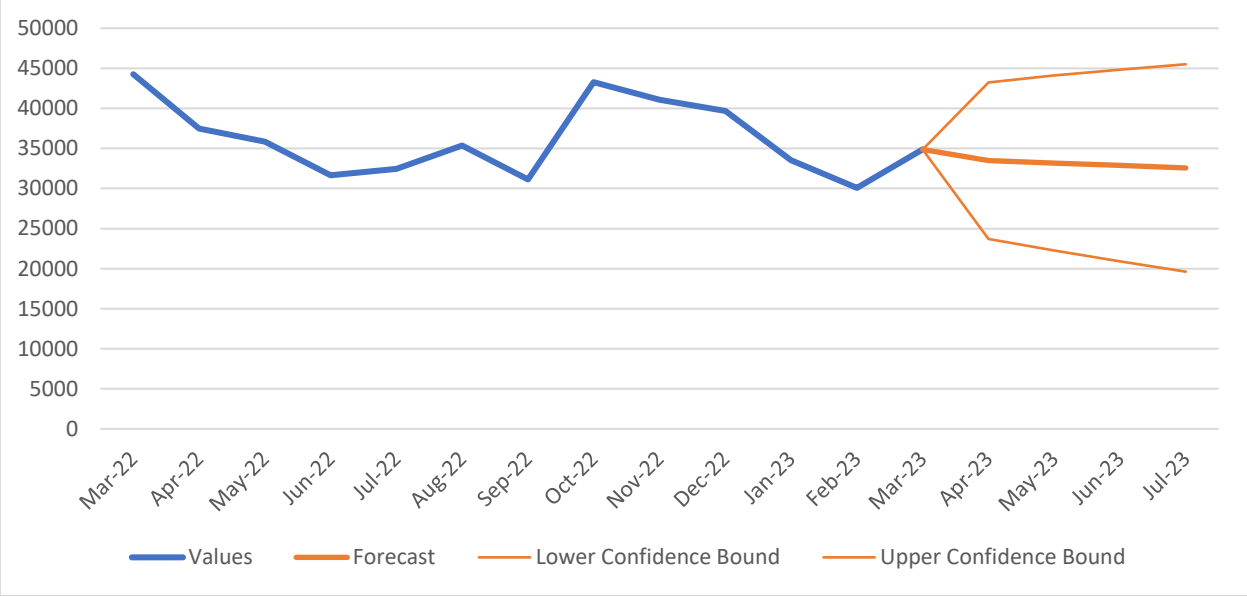
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -460.52100    8.97227  -51.33   <2e-16 ***
price_unit     8.05712    0.09758   82.57   <2e-16 ***
quantity      61.26698    0.88323   69.37   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 6824.636)

    Null deviance: 87151479  on 1029  degrees of freedom
Residual deviance:  7008901  on 1027  degrees of freedom
AIC: 12021

Number of Fisher Scoring iterations: 2
```

At, last we forecasted the total sales for the Departmental Store and we conclude that the sales will be more comparatively to the previous years/months.



(DECISION TREE, KNN, RANDOM FOREST)

```
# decision tree
set.seed(7)
fit.cart <- train(total~., data=store_cleaned_2, method="rpart",metric=metric, trControl=control)
# Random Forest
set.seed(7)
fit.rf <- train(total~., data=store_cleaned_2, method="rf",metric=metric, trControl=control)
# kNN
set.seed(7)
fit.knn <- train(total~., data=store_cleaned_2, method="knn",metric=metric, trControl=control)
```

```
> #making prediction
> predict(fit.cart)
       1        2        3        4        5        6        7        8        9       10
819.9027 198.1983 430.4385 819.9027 819.9027 819.9027 819.9027 819.9027 198.1983 198.1983
      11       12       13       14       15       16       17       18       19       20
198.1983 198.1983 430.4385 819.9027 819.9027 819.9027 819.9027 819.9027 198.1983 198.1983
      21       22       23       24       25       26       27       28       29       30
819.9027 430.4385 198.1983 198.1983 430.4385 819.9027 198.1983 430.4385 819.9027 198.1983
      31       32       33       34       35       36       37       38       39       40
819.9027 819.9027 819.9027 430.4385 819.9027 198.1983 430.4385 819.9027 819.9027 430.4385
      41       42       43       44       45       46       47       48       49       50
198.1983 198.1983 819.9027 819.9027 198.1983 819.9027 819.9027 198.1983 198.1983 819.9027
      51       52       53       54       55       56       57       58       59       60
819.9027 430.4385 198.1983 198.1983 198.1983 819.9027 198.1983 819.9027 819.9027 198.1983
      61       62       63       64       65       66       67       68       69       70
198.1983 430.4385 819.9027 198.1983 430.4385 198.1983 198.1983 819.9027 819.9027 198.1983
      71       72       73       74       75       76       77       78       79       80
819.9027 819.9027 198.1983 819.9027 819.9027 430.4385 819.9027 198.1983 819.9027 198.1983
      81       82       83       84       85       86       87       88       89       90
819.9027 430.4385 198.1983 430.4385 430.4385 819.9027 819.9027 430.4385 198.1983 819.9027
      91       92       93       94       95       96       97       98       99      100
198.1983 819.9027 198.1983 819.9027 198.1983 430.4385 819.9027 198.1983 198.1983 819.9027
     101      102      103      104      105      106      107      108      109      110
198.1983 819.9027 198.1983 430.4385 198.1983 819.9027 198.1983 430.4385 430.4385 819.9027
     111      112      113      114      115      116      117      118      119      120
198.1983 430.4385 819.9027 819.9027 819.9027 198.1983 198.1983 198.1983 198.1983 198.1983
     121      122      123      124      125      126      127      128      129      130
819.9027 819.9027 819.9027 819.9027 819.9027 819.9027 198.1983 430.4385 819.9027 819.9027
     131      132      133      134      135      136      137      138      139      140
430.4385 819.9027 198.1983 819.9027 819.9027 430.4385 198.1983 198.1983 819.9027 819.9027
```

```
> predict(fit.rf)
       1        2        3        4        5        6        7        8
730.07820 123.03561 456.23548 635.24410 841.17367 832.91370 599.34044 959.43503
       9       10       11       12       13       14       15       16
160.03268 299.35737 100.63379 171.38794 360.24557 569.22543 931.17328 812.35693
      17       18       19       20       21       22       23       24
676.29240 631.60265 298.74157 177.15389 652.25048 476.75210 146.85743 267.22795
      25       26       27       28       29       30       31       32
480.32994 574.87858 111.91440 380.49590 668.83399 306.46862 713.20021 931.60870
      33       34       35       36       37       38       39       40
910.23905 418.68160 645.37424 222.21983 477.66408 730.07721 599.60547 335.34837
      41       42       43       44       45       46       47       48
282.33064 245.10983 601.08427 1068.30047  70.81696 610.81949 680.78044 249.15947
      49       50       51       52       53       54       55       56
171.81430 1076.43001 892.21198 343.00053 122.13071  52.74138  73.83950 931.79364
      57       58       59       60       61       62       63       64
194.33487 970.66819 941.97524 271.90175 138.75907 486.83689 661.68875 217.84929
      65       66       67       68       69       70       71       72
493.46130 218.35698 148.02581 845.14699 1025.90168  62.43043 1162.55414 812.39434
      73       74       75       76       77       78       79       80
265.63481 659.99749 892.07288 549.33700 589.94739 249.14536 1020.51119 161.13312
      81       82       83       84       85       86       87       88
859.31517 523.71264 218.35530 523.38456 374.78009 811.60406 580.97825 486.80517
      89       90       91       92       93       94       95       96
140.09176 668.61846 311.64981 590.18032 222.46462 770.69085 291.54652 315.37214
      97       98       99      100      101      102      103      104
1140.89482 116.24126 288.05004 717.85252 160.07927 939.68230 285.04577 508.37576
     105      106      107      108      109      110      111      112
303.49063 1156.38543 244.34238 541.59116 413.49583 1067.70733  75.35439 532.06652
     113      114      115      116      117      118      119      120
712.99622 696.38840 963.73127 150.78229 173.12342 168.63405 156.18318 233.70209

> predict(fit.knn)
  [1]  733.79316  104.65884  459.61422  633.12328  841.25326  862.91862  601.11316
  [8]  936.78928  155.20778  280.64388  107.44298  160.90232  354.39256  559.00652
 [15]  936.78928  795.69052  678.87648  627.76258  282.79886  172.38984  688.57496
 [22]  468.06294  140.25988  273.37644  471.89782  570.78936  108.17486  401.41906
 [29]  679.30020  302.95980  694.88796  909.47218  920.68578  419.25596  655.96350
 [36]  236.40580  497.32530  711.29534  605.14706  322.51084  281.99208  248.66158
 [43]  592.53604 1034.55090   74.52764  623.50184  677.81290  253.35674  168.14194
 [50] 1057.33120  906.62812  337.61496  117.99104   90.61402   73.45122  924.24460
 [57]  200.53726  955.60844  931.82234  272.53542  130.63844  497.82392  661.78216
 [64]  228.87086  506.38606  218.46618  140.39684  863.29740 1034.55090   60.66258
 [71] 1188.36126  810.64698  264.69660  646.66734  893.90582  560.16854  599.41828
 [78]  253.14060 1005.11734  150.96202  842.03864  516.58530  219.47840  552.57154
 [85]  368.99592  804.59506  585.32852  489.17618  137.33664  671.06120  298.52358
 [92]  588.20468  217.51602  765.59784  288.21734  303.85646 1142.20360  120.81584
 [99]  269.65284  729.41686  181.46344  920.91262  276.87320  502.24944  277.95390
[106] 1142.20360  244.20182  535.75756  423.39258 1109.40168   72.77498  521.42812
[113]  704.44948  690.59940  951.11658  155.22490  170.51948  176.37238  150.82506
[120]  239.76774 1078.60494  556.46634 1191.49208  703.93160  595.39080  908.67824
[127]  253.63494  385.58092  745.05384 1042.55878  389.01348  791.76362  227.76448
[134]  757.44230  717.03696  493.71726  205.06336  281.17032  714.88626  765.59784
[141] 1130.38224 1130.38224  894.94586  109.67928  663.23094  424.84778  395.52122
[148]  423.27488  777.87288  353.42528  603.82454  356.19016 1014.32790  148.35550
[155]  862.47564  714.88626  250.75878  392.61724 1128.44126  791.04458  489.17618
[162]  226.88922  447.13802  313.02422  532.88354  450.77816 1184.86878 1235.25080
[169]  444.80756  667.75062  521.42812  609.17240  248.20790  453.53876  232.08514
[176]  399.94460  270.17928  228.26524  633.01414  468.55300  623.38414  434.67680
[183]  197.25450  380.17314  130.38806  116.45880 1032.71050  289.99140  237.30888
[190]  444.03074  424.52464  308.85336 1009.09774  171.05876  261.94670  245.89242
[197]  192.89746   65.08596  275.71760  541.72602  171.27276  379.67238  581.52146
[204]  321.31886  135.38068  282.98290  773.96096  370.54100  176.67626 1177.15408
```

```
> results <- resamples(list(cart=fit.cart, knn=fit.knn, rf=fit.rf))#,nn =fit.nn))
> summary(results)

Call:
summary.resamples(object = results)

Models: cart, knn, rf
Number of resamples: 10

MAE
          Min.   1st Qu.    Median       Mean   3rd Qu.       Max. NA's
cart 65.1076956 67.336169 70.136847 79.8187524 95.822552 100.574657    0
knn  13.7863371 15.708224 18.087232 17.9423637 19.796767  23.113893    0
rf    0.7404898  0.903209  0.973188  0.9690696  1.026963   1.326397    0

RMSE
         Min.  1st Qu.    Median      Mean    3rd Qu.       Max. NA's
cart 75.324307 78.74848 84.463459 98.777879 125.025588 130.587107    0
knn  19.700094 22.14168 29.335759 29.013972  33.913102  40.057111    0
rf    1.115165  1.62466  1.738099  1.859475   2.184003   2.728679    0

Rsquared
         Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
cart 0.8062919 0.8168307 0.9214544 0.8822198 0.9283444 0.9372137    0
knn  0.9830740 0.9874281 0.9911988 0.9906426 0.9936928 0.9956826    0
rf   0.9999213 0.9999393 0.9999639 0.9999575 0.9999720 0.9999850    0

> #making prediction
```

Conclusion-

- As a business owner, based on our Analysis we conclude that the sales for the departmental store will eventually increase and as there is recession going on we believe that the demand and supply for the Goods and Services will be at large.
- Further, we suggest the Departmental Store to try focus on marketing the products and shipping and distributing the same equally across all the Locations.
- The Departmental Store should also focus on Hiring the Employees and also try to make changes to the existing products by segregating the same products according to the demands of the public.
- At last, we conclude that the Departmental Store will run good business once they start targeting the Customer policies and concentrate on increase sales of different categories.